Introduction: From Historical GIS to Spatial Humanities: Deepening Scholarship and Broadening Technology

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WHEN GEOGRAPHICAL INFORMATION SYSTEMS (GIS) FIRST began to be used by academic geographers in the late 1980s, their use was nothing if not controversial. Proponents of the new field argued that it had the potential to reinvigorate geography as a discipline under a more computational paradigm.[1] Opponents argued that it marked a lurch toward an unacceptable form of positivism with no epistemology or treatment of ethical or political issues.[2] One thing on which they both agreed – or perhaps took for granted – was that GIS was a quantitative technology that was to be used in a social scientific manner (to its supporters) or a positivist way (to its antagonists).

When GIS first began to be used by historians it was not surprising that much of the early focus was also quantitative and social science based. It is no coincidence that the first special issue of a journal dedicated to historical GIS (HGIS), published in *Social Science History,* included essays on topics such as fertility, migration, urban history, and economic growth, all well suited to quantitative analysis.[3] In 2008, eight years after this issue was published, a conference devoted to HGIS was held at the University of Essex.[4] It attracted 125 delegates, with papers organized in 21 sessions. While some of these sessions were themed on topics that still had a strong quantitative bent – demography, urban history, environmental history, transport, and so on – there was also an increasing number of papers and sessions that concentrated on topics that were clearly qualitative and did not follow traditional social science paradigms. These topics included art, performance culture, literature, the Bible, and medieval and early modern history. What was happening in the quantitative sessions was also interesting. Rather than concentrating on issues associated with database construction and potential applications, many of these papers had developed to focus on conducting applied works of history – studies that developed the historiography by answering applied research questions. This was an indication of two emerging trends within HGIS that have continued since: HGIS is deepening from an applied perspective, and it is broadening from a technical perspective. It is deepening in that it has reached a stage where researchers apply it to scholarship that develops new knowledge about the past. This must be the ultimate aim of the field, as it takes HGIS beyond a narrow technical specialism and makes it relevant to a much wider audience. HGIS is also broadening its technical scope in terms of the ever-widening potential for its application to both qualitative and quantitative sources. This means that GIS is thus able to expand beyond social science history – a fairly narrow field – to be applicable to the discipline more broadly, and beyond that to spread outside the disciplinary boundaries of history into other humanities disciplines.

GIS AND HISTORICAL RESEARCH

There are many different definitions of GIS and related terms such as GISc (Geographical Information Science).[5] The emergence of new geospatial technologies such as Google Earth that do not fit traditional definitions only complicates these definitions. Originally, "GIS" was considered as the umbrella term for the field, and it is often still used in this way. The more recent trend, however, has been to use "GIS" to describe the tools offered, while "GISc" emphasizes the broader understanding of how these tools can be developed, used, and applied.[6]

To take this further, GIS can be thought of as a type of software that provides a way of representing features on the Earth's surface and a suite of operations that allow the researcher to query, manipulate, visualize, and analyze these representations. The representations, or data models, combine two types of data: *attribute data,* which were traditionally held in a table and tend – or perhaps tended – to be quantitative, and *spatial data,* which locate each item of data using a point, a line, a polygon (which represents an area or a zone), or a pixel. Points, lines, and polygons are used to represent discrete features, and data in these formats are referred to as *vector data,* while pixels are used to represent continuous surfaces and are referred to as *raster data.*[7] In this way the attribute data say *what,* while the spatial data say *where.* Thus, from this perspective GIS is a type of software that allows the user to store, retrieve, visualize, and analyze data that are *georeferenced* to a location on the Earth's surface.[8] GIS allows researchers to ask questions about their topics or sources that stress the importance of location and thus geography. This emphasis on geography, combined with the tools to represent and explore georeferenced data, is what allows scholars to conduct their research in new ways.

This approach to defining GIS leads to the conclusion that a Geographic Information System is really a database for managing georeferenced data. Until recently this conclusion made a quantitative paradigm almost inevitable, as databases were, almost by definition, quantitative, holding either numbers or structured textual information such as is found in library catalogs. Recent developments in Information Technology (IT) mean that this paradigm is no longer the case. Increasingly, almost any type of data can be held within a computer system, including unstructured texts such as books and web pages, still images, moving images, and sounds. As long as a location can be found for these items, they can be held within a GIS-type structure. This means that the need for attribute data to be quantitative is increasingly disappearing.

While this is true of attribute data, it is not true of spatial data, which, although they tend to be represented graphically, are indisputably quantitative in nature. What appears to be a point on the map is actually a pair of numbers representing *x* and *y* or latitude and longitude. A line is a series of points joined together, and the boundaries of a polygon are made up of one or more lines. Despite the many critiques of maps, researchers are usually far more comfortable interpreting the crude quantitative abstraction of space created from spatial data than they have been interpreting the crude quantitative abstraction of society created from quantitative data.[9] For example, many humanities researchers would be happy with a map showing the locations of certain events as points but would be suspicious of a scatterplot showing the relationship between two variables such as the unemployment rate and the number of crimes in tracts around a city. In reality, the two have much in common. They both simply show dots whose location is determined using values of *x* and *y,* values that are typically expressed to a far higher degree of precision than the accuracy of their measurement can really support.[10] There are a number of possible reasons for this apparent inconsistency. These include the valid reason that measuring and interpreting statistics about characteristics of the population are more difficult than doing the same for locations on the Earth's surface. Less justifiable are a misplaced confidence in the authority of maps and, from the opposite perspective, a misplaced suspicion of quantitative approaches among some researchers in the humanities.

Using GIS in humanities research presents the researcher with two major sets of challenges. The first is to get the data into a GIS. GIS databases are usually vector data, which require that every item of attribute data be located using a precisely defined point, line, or polygon. Techniques such as using raster surfaces or networks have been used effectively in HGIS research to represent imprecision in location, but even these approaches usually require a precise point-based location to be given initially and do not solve the inherent uncertainties within the spatial data.[11] In some cases it may simply be impossible to get a source into a form suitable for GIS; however, as we will see, this does not mean that such a source cannot contribute to a study that is centered on a GIS database.

The second, and more important, challenge is to get information back from the GIS databases and turn it into new scholarship that advances our knowledge of the past. Once a GIS database has been created it is very easy to produce large numbers of maps, graphs, tables, and summaries. Going beyond this to produce new knowledge or an innovative narrative requires a different set of skills. Creating a GIS and analyzing the data that it contains requires technical GIS skills. Producing new scholarship requires the skills of the historian or other humanities scholar to turn the GIS output into a contribution to our understanding of the past. A key test of the effectiveness of historical research is that the work that it produces should be of interest not only to HGIS specialists but also to an audience of subject specialists who are more interested in the results of the research than in the methodology that was used to achieve these results.

These challenges show that there are certain principles that researchers interested in GIS must consider. Clearly, there must be a geographical dimension to whatever study is being undertaken, and, beyond this, the requirements and limitations of the vector and raster data models need to be understood. However, the fact that HGIS research is ultimately based on a software tool does not, as is sometimes claimed, force a positivist approach onto the researcher. As described above, GIS merely provides a platform on which research can be conducted. It does not impose any approach other than the fact that the data within the GIS database have to be represented using attribute and spatial data and that the spatial data must be in the form of points, lines, polygons, or pixels. How scholars turn these data into information about the past and then to humanities scholarship is their decision. They would start from their discipline's existing paradigms and add the more explicitly geographical. That said, as noted above, GIS does require locations that are usually expressed with high degrees of precision – usually far more precision than it makes sense to express them to. This is not, of itself, positivism, which is concerned with using statistical approaches to define relationships between variables so that empirical generalizations can be made.[12] Early fears of GIS causing a return to "the very worst sort of positivism" were prompted by calls for GIS to reinvigorate the use of quantitative attribute data and statistical approaches rather than by fears about the accuracy of spatial data.[13] It is also worth noting that representing the location of a mountaintop or a city using a point whose coordinates are given submillimetric precision makes no more sense in the earth sciences or social sciences than it does in the humanities. The limitations of the vector data model mean that locations are expressed with this spurious level of precision. It is up to the researcher to interpret this precision and its consequences sensibly, which is not always as easy as it may seem; indeed, here the humanities may have an advantage over more empirical approaches. While in other fields coordinates are often used as inputs into statistical approaches where their overly precise nature becomes lost in summary statistics, in the humanities – where the emphasis is on close reading and careful interpretation – imprecision and ambiguity are actually easier to handle, as these limitations remain more transparent to the critical researcher.

It is not the intention here to discuss existing humanities paradigms, but it is worth discussing in general terms what adding the use of GIS has to offer to them. Traditionally, it has been argued that there are three main advantages in using GIS in historical research: GIS structures the data to allow them to be discovered and explored in ways that are explicitly spatial; it allows the data to be visualized using mapping and other approaches; and it allows the data to be analyzed in ways that are explicitly spatial.[14] A fourth advantage, and one whose importance is frequently underestimated, is the ability of GIS to integrate data from a wide range of apparently incompatible sources. As all of the data are georeferenced to specific coordinate-based locations on the Earth's surface, at a technical level at least, any dataset can be integrated with any other dataset to see how the locations within one dataset compare with the locations in another. This integration has potentially major benefits, as, for example, previously disparate and apparently incompatible sources can be brought together.

Thus GIS can be thought of as a tool that enables researchers to explicitly handle space and location. It is far from a perfect tool, as its data models are crude. It is, however, a highly effective tool with much to offer many subjects across the humanities as long as its limitations are understood and the patterns that it reveals are evaluated critically.

TRENDS IN HGIS

As mentioned above, the 2008 conference illustrated that HGIS was being taken in two directions. On the one hand, the more traditional side of the field – the quantitative, social science–based side – was moving away from its original technical emphasis to focus increasingly on answering research questions and developing new narratives. In this respect the field was deepening as it moved from the technical to the applied. This change is reflected in the fact that the term "historical GIS" – with its clear emphasis on technology – is increasingly being replaced with the term "spatial history," an expression that stresses doing a form of history that emphasizes geography.[15] A key point about this is that as research within the field has developed, it has become increasingly topic based rather than technology or data based. This deepening of the field also represents a widening and a move toward maturity, because the new results lead to it being of interest to a broad audience of historians.

At the same time as this deepening, the field is also broadening at a technical level to allow it to address a greater range of sources; to explore, analyze, and disseminate them in new ways; to develop new questions that could not previously be asked; and to move into new subject areas. This broadening is happening on the quantitative side; it also, to an increasing extent, has developed the use of GIS into qualitative sources. This exciting development means that, rather than concentrating on social science history, the field is broadening into history more generally and also into other humanities disciplines. At present the emphasis is still on technology, data infrastructure, and potential. This is understandable and does not represent a major criticism. All GIS projects experience long lead times as databases are built. Investigations into what the data and technology are capable of offering then have to be made before applied research can take place. The social science end of HGIS went through this process for a number of years before it began to turn into the more applied field of spatial history. The broadening of HGIS to include qualitative sources is leading to the development of "humanities GIS," a field whose techniques and approaches have the potential to be applied across the humanities. This in turn provides a foundation for "spatial humanities," a field using geographical technologies to develop new knowledge about the geographies of human cultures past and present.[16]

THE ESSAYS IN THIS VOLUME

In this volume we provide six essays that showcase the deepening and broadening trends discussed above. The volume is divided into two parts of three essays each. The first part focuses on the deepening of the field into the applied scholarship that develops historiography – the move from HGIS to spatial history. It includes three essays that are based on large quantitative HGIS databases but that conduct applied research on a variety of very different topics. Robert Schwartz and Thomas Thevenin explore how agricultural change in England and Wales and in France was affected by the development of the rail network. Andrew Beveridge explores the changing patterns of segregation in U.S. cities over the long term, and Niall Cunningham explores a variety of questions associated with long-term religious change in Ireland and the violence that has sometimes accompanied it.

The second set of three essays explores broadening the technology into new areas. Humphrey Southall explores a variety of ways in which the Great Britain Historical GIS can be applied beyond the traditional boundaries of history. There are similarities between his essay and the one by Cunningham that precedes it in that both explore the potential of large HGIS databases for Ireland and Britain, respectively. The major contrast is that while Cunningham's essay concentrates very much on social science history approaches, Southall approaches the topic far more broadly and explores themes as diverse as modern medical demography, environmental change, and commercial applications. The other two essays in this part are more firmly based on qualitative sources and the shift from traditional historical GIS toward humanities GIS. Elijah Meeks and Ruth Mostern look at a range of potential uses for a major gazetteer of places in Song dynasty China. Julia Hallam and Les Roberts present a much more focused essay that explores the potential for the use of an archive of amateur films to help historians understand the city of Liverpool in the 1950s. Each essay is described in more detail at the start of each part.

Together, the six essays cover a broad range of subjects and scales – ancient and modern, national and local, rural and urban – and cover the spectrum from topic-based work that answers specific research questions to source-led or data-led work that is concerned with the development of new approaches and their potential applications. There are, however, some key themes that run through them all. These are particularly associated with the fact that GIS allows historians to make extensive use of the geographical nature of their sources. This goes well beyond simple mapping. As stated above, one of the key advantages of GIS is that it allows data from disparate sources to be integrated. Schwartz and Thevenin integrate agricultural statistics with data on the transport network. Southall integrates census data and a wide range of other sources. Cunningham integrates multiple censuses for Ireland to explore change over time and also integrates this polygon-based census information with a major database of killings during Northern Ireland's Troubles to allow violent deaths to be compared with background social and economic variables. All of the authors show the importance of applying geography and location to their research topics, but all of the essays are based on a wide range of different approaches to history. These stretch from Beveridge's highly quantitative approach to the study of segregation in U.S. cities based on census data to Hallam and Roberts's study of films of Liverpool. Finally, as discussed above, all of the essays are based on adding GIS approaches to existing paradigms in topics as diverse as Schwartz and Thevenin's study of Victorian Europe and Meeks and Mostern's study of Song dynasty China.

GIS is therefore a technology that provides scholars with a tool to assist them with their research. It is not a tool that forces any particular academic paradigm onto researchers; indeed, as the essays in this book show, GIS can be used with a wide variety of different approaches to different topics. It is a tool that relies on researchers being able to represent their data in a particular way based around linking attribute information about locations to precisely represented spatial data, particularly points, lines, and polygons. Not all data can be represented in this way; therefore, not all data can be explicitly incorporated into a GIS-based analysis. A study that has GIS at its core, however, does not need to exclude other types of evidence. As a tool, GIS clearly encourages researchers to think about location and geography. As soon as researchers create their database, they will map it, and much of the subsequent research will involve manipulating, refining, enhancing, and interpreting the maps that the database produces.

However, maps rarely answer questions; far more commonly, they pose them.[17] Why is the pattern as it is? Why are things different over here compared to over there? It is up to researchers to answer these questions in a way that they choose – GIS does not force a paradigm onto them. This presents a challenge. The technology was developed for reasons that have little to do with the needs of academic researchers, particularly those in the humanities. The challenge for humanities researchers is to take these tools and modify, develop, and apply them in ways that are appropriate to the paradigm that they want to pursue. As the subsequent essays show, this process is not always easy, but it does provide new and exciting opportunities to develop new knowledge in a wide range of disciplines and topics that focus on the study of geographies of the past.

NOTES

1. See, in particular, the work of S. Openshaw, "Towards a More Computationally Minded Scientific Human Geography," *Environment and Planning A* 30 (1998): 317–32; and S. Openshaw, "A View on the GIS Crisis in Geography, or, Using GIS to Put Humpty-Dumpty Back Together Again," *Environment and Planning A* 23 (1991): 621–28.

2. See P. J. Taylor, "Editorial Comment: GKS," *Political Geography Quarterly* 9 (1990): 211–12; P. J. Taylor and M. Overton, "Further Thoughts on Geography and GIS – a Pre-emptive Strike," *Environment and Planning A* 23 (1991): 1087–90; and the essays in J. J. Pickles, ed., *Ground Truth: The Social Implications of Geographic Information Systems* (New York: Guildford, 1995).

3. G. W. Skinner, M. Henderson, and Y. Jianhua, "China's Fertility Transition through Regional Space," 613–52; I. Gregory, "Longitudinal Analysis of Age- and Gender-Specific Migration Patterns in England and Wales," 471–503; L. Siebert, "Using GIS to Document, Visualize, and Interpret Tokyo's Spatial History," 537–74; R. G. Healey and T. R. Stamp, "Historical GIS as a Foundation for the Analysis of Regional Economic Growth," 575–612, all in "Historical GIS: The Spatial Turn in Social Science History," ed. A. K. Knowles, special issue, *Social Science History* 24, no. 3 (2000).

4. See http://www.hgis.org.uk/HGIS_conference/2008/index2008.htm for details.

5. N. R. Chrisman, "What Does 'GIS' Mean?," *Transactions in GIS* 3 (1999): 175–86.

6. M. F. Goodchild, "Geographical Information Science," *International Journal of Geographical Information Systems* 6 (1992): 31–45; D. J. Wright, M. F. Goodchild, and J. D. Proctor, "Demystifying the Persistent Ambiguity of GIS as 'Tool' versus 'Science,'" *Annals of the Association of American Geographers* 87 (1997): 346–62.

7. Any good introductory textbook to GIS will discuss these definitions in detail. See, for example, N. Chrisman, *Exploring Geographic Information Systems,* 2nd ed. (New York: John Wiley, 2002); F. Harvey, *A Primer of GIS: Fundamental Geographic and Cartographic Concepts* (New York: Guildford, 2008); I. Heywood, S. Cornelius, and S. Carver, *An Introduction to Geographical Information Systems,* 4th ed. (Harlow, Essex: Prentice Hall, 2012); D. Martin, *Geographic Information Systems and Their Socio-economic Applications,* 2nd ed. (Hampshire: Routledge, 1996).

8. See, for example, Department of the Environment, *Handling Geographical Information: Report of the Committee of Enquiry Chaired by Lord Chorley* (London: HMSO, 1987); D. J. Maguire, "An Overview and Definition of GIS," in *Geographical Information Systems: Overview, Principles, and Applications,* ed. D. J. Maguire, M. F. Goodchild, and D. W. Rhind (Harlow, Essex: Longman, 1991), 9–20; and D. F. Marble, "Geographical Information

Systems: An Overview," in *Basic Readings in GIS,* ed. D. Peuquet and D. F. Marble (London: Taylor & Francis, 1990), 8–17.

9. For examples of critiques of maps, see M. Monmonier, *How to Lie with Maps,* 2nd ed. (Chicago: University of Chicago Press, 1996); J. B. Harley, "Cartography, Ethics and Social Theory," *Cartographica* 27 (1990): 1–23; J. Pickles, *A History of Spaces: Cartographic Reason, Mapping and the Geo-coded World* (London: Routledge, 2003).

10. I. N. Gregory, "'A Map Is Just a Bad Graph': Why Spatial Statistics Are Important in Historical GIS," in *Placing History: How Maps, Spatial Data and GIS Are Changing Historical Scholarship,* ed. A. K. Knowles (Redlands, Calif.: ESRI Press, 2008), 123–49.

11. For raster surfaces, see, for example, K. Bartley and B. Campbell, "*Inquisitions Post Mortem,* GIS and the Creation of a Land-Use Map of Medieval England," *Transactions of GIS* 2 (1997): 333–46; and D. Cooper and I. N. Gregory, "Mapping the English Lake District: A Literary GIS," *Transactions of the Institute of British Geographers* 36 (2011): 89–108. For an example of networks, see M. L. Berman, "Boundaries or Networks in Historical GIS: Concepts of Measuring Space and Administrative Geography in Chinese History," *Historical Geography* 33 (2005): 118–33.

12. See R. J. Johnston, *Philosophy and Human Geography: An Introduction to Contemporary Approaches* (London: Edward Arnold, 1983), chap. 2; and A. Holt-Jensen, *Geography: History and Concepts,* 2nd ed. (London: Paul Chapman, 1988), chap. 4.

13. The quote is from Taylor, "Editorial Comment," 211.

14. I. N. Gregory, K. K. Kemp, and R. Mostern, "Geographical Information and Historical Research: Current Progress and Future Directions," *History and Computing* 13 (2003): 7–24.

15. R. White, "What Is Spatial History?," http://www.stanford.edu/group/spatialhistory/cgi-bin/site/pub.php?id=29.

16. D. J. Bodenhamer, "The Potential of Spatial Humanities," in *Spatial Humanities: GIS and the Future of Humanities Scholarship,* ed. D. J. Bodenhamer, J. Corrigan, and T. M. Harris (Bloomington: Indiana University Press, 2010), 14–30.

17. See A. R. H. Baker, *Geography and History: Bridging the Divide* (Cambridge: Cambridge University Press, 2003), or, for a more applied example of how geographical thinking was far more important than a single map in John Snow's famous work on cholera in Victorian London, see S. Johnson, *The Ghost Map: A Street, an Epidemic and the Hidden Power of Urban Networks* (Penguin: London, 2006).

Toward Spatial Humanities

SIX

Mapping the City in Film

JULIA HALLAM AND LES ROBERTS

IN THIS CHAPTER WE EXAMINE HOW GEOSPATIAL COMPUTING tools such as GIS can contribute to an understanding of the development of local film culture and its contribution to projections of "place," drawing on archival research into Liverpool and Merseyside on film. We will map some of the contradictory and ambiguous spatialities that historically have mediated ideas of "the local" and "the regional" in a range of moving image genres, exploring the correlations between categories of genre, date, and location as assessed in relation to records in a spatial database consisting of over seventeen hundred films shot in Merseyside from 1897 to the 1980s. Significantly, the use of GIS has revealed the ways in which particular styles and genres of filmmaking create their own cinematic maps, initiating new modes of spatial dialogue between the virtual landscapes of the moving image and the architectural, geographic, and imagined spaces within which they are embedded.

A provincial city on the Mersey estuary in England's northwest of around four hundred thousand people, Liverpool is internationally renowned for its football teams (Liverpool and Everton), its music (the 1960s Mersey sound and the Beatles), its infamous slave-trading past, and the three buildings at the Pier Head that dominate its iconic waterfront – the Royal Liver, Cunard, and Port Authority buildings, colloquially known as the "Three Graces." Granted UNESCO world heritage status in 2004 for its innovative enclosed dock systems and grand nineteenth-century neoclassical civic buildings, the once-thriving port, deemed in the nineteenth century the "second city" of the British Empire, has reinvented itself for the twenty-first century as a postindustrial city dependent, at least in part, on heritage and cultural tourism for its continuing economic development and prosperity. The city shared this pattern of growth, decline, and regeneration with many port cities in Europe and North America during the nineteenth and twentieth centuries, their fortunes ebbing and flowing with the shifting tides of global capital and commercial trade.

The films collated in the City in Film database document urban life throughout a century of decline, redevelopment, and reinvention. Visitors passing through the city in the 1890s, many of them European migrants on their way to a new life in the United States aboard the liners that plied the Atlantic trade route between Liverpool and New York, could marvel at the nine miles of busy docklands by traveling the length of the waterfront on the first electric overhead railway in the world. This was a journey taken by Jean Alexandre Promio, a cameraman working for the Lumière Brothers who recorded the first moving images of the docks from one of the railway carriages in one of the first known instances of a "tracking" shot (*Panorama pris du chemin de fer électrique,* Lumière Brothers, 1897). Promio also recorded the modern, architecturally innovative office blocks in the commercial district around the Strand, Water Street, Castle Street, and Dale Street, capturing the busy shoppers on adjacent Lord Street and Church Street (*Church Street,* Lumière Brothers, 1897) and the vestments of civic pride enshrined in the grandeur of St. George's Hall, shot from St. George's Plateau, just outside Lime Street railway station (*Lime Street,* Lumière Brothers, 1897). Based on a reading of the length of the shadows in Promio's images, it seems probable that, like many visitors to the city, Promio arrived at Lime Street and recorded his images, crossing St. George's Plateau as he traveled from the station to the waterfront via the main thoroughfare, Church Street.[1]

Today, the neoclassical hall and the accompanying buildings on the Plateau (the Walker Art Gallery, the William Brown and Picton Libraries in tandem with the Albert Dock, and the Three Graces at the Pier Head) form the core of the maritime mercantile city. Unusually, the city boasts two cathedrals, both built in the twentieth century. The imposing Anglican Cathedral, designed by Giles Gilbert Scott in 1903 in the Gothic Revival style, is the fifth largest cathedral in the world. Towering above the commercial district and Liverpool's Chinatown on St. James's Mount, it took seventy-four years to build and was finally completed in 1978. Facing it, close to the University of Liverpool's Victoria Building at the other end of Hope Street, is the Metropolitan Cathedral of Christ the King, a modernist building affectionately known locally as "Paddy's wigwam," built in the 1960s. Investment in the 1960s sought to improve road access in and around the city by constructing a second Mersey tunnel in the midst of the close-knit, inner-city, working-class district of Scotland Road, close to the first tunnel entrance (opened in 1934) behind St. George's Hall at the bottom of William Brown Street. Other developments included a modern indoor shopping complex opposite Lime Station on the site of the old fruit and vegetable market; the imposing tower, topped by a revolving restaurant, now forms part of Liverpool's iconic skyline. The twenty-first century has witnessed the building of a new shopping and leisure complex, Liverpool One, opposite the Albert Docks on the site of the old Customs House (bombed during the Second World War), the development of an arena and conference center on the former Kings Dock, the opening of a new cruise ship terminal at the Pier Head, and the construction of the largest newly built national museum in the UK for over a hundred years, the Museum of Liverpool.[2]

Films made in and about the city focus on many of these developments, charting changes in the urban landscape and the effects of these often controversial regeneration schemes on the city's many and varied communities. The mapping database has been developed in part with museum curators who, inspired by the City in Film project, have begun the task of georeferencing and digitizing materials and artifacts relating to the Merseyside area in the collections of National Museums Liverpool. These are being housed in a permanent electronic resource modeled on a GIS database in the new Museum of Liverpool "history detectives" gallery (opened in 2011), which enables public access via a map-based touch-screen interface to images, films, and audio that begin to reveal the distinctive histories and identities of the people and places of the Mersey region.

FILM MAPPINGS

Intellectually, the project draws on and develops from work into the relationship between film and place, a relationship that has principally focused on the city as a space in which the activities of filmmaking and filmgoing form a nexus of enquiry informed by interdisciplinary and multidisciplinary perspectives. Following the "spatial turn" in the humanities and social sciences in the 1990s, the term "mapping" has gathered significance.[3] A growing vanguard of researchers is studying the relationship between film, space, and place from disciplines that range from geography, urban studies, architecture, and history to literature, film, media, and cultural studies. What motivates much of the work across this apparently disparate field is an interest in the ways in which the interdisciplinary study of moving images, and the cultures of distribution and consumption that develop in tandem with the production of those images, provides renewed insights into our knowledge of the development of urban modernity and modern subjectivity. In his survey of the emerging field of what may loosely be termed "cinematic cartography," Les Roberts explores the different ways in which "thinking spatially" underpins a growing body of research on film and the moving image, noting that it is becoming increasingly difficult to gauge what is meant by the spatial turn and the ubiquitous trope of mapping, which is found in much contemporary cultural criticism.[4] Echoing Henri Lefebvre's argument that space is culturally and materially reproduced as part of lived everyday reality, Roberts argues that there is a need to situate and embed visual cultures in social and material landscapes and to explore or "map" the interplay between the representational and the material.[5] In film studies, for example, although there has been theoretical concern with genealogical mappings of the discursive terrain that created the conditions that we now know as cinema, the dynamic and dialectical interplay between different generic spatial formations and their role in the material and symbolic production of social space is less well explored.[6] Responding to Giuliana Bruno's suggestion that film is a form of "modern cartography," Roberts identifies some of the different ways that film and cartography have begun to find convergence, theoretically, methodologically, and practically, creating a number of strands in the emerging field of what can be broadly defined as "cinematic cartography."[7] These are briefly summarized in the following five paragraphs.

The first of these categories and the one that is most securely grounded in the material and symbolic aspects of place is the mapping of film production and reception. In the United States, scholars such as Robert C. Allen and Jeffrey Klenotic have begun to explore the use of Geographical Information Systems (GIS) and digital mapping in historical studies of film distribution and consumption. As Allen points out, despite the "historical turn" that has shaped recent directions in film scholarship, as a discipline "film studies continues to be dogged by ambivalence towards the use of empirical methods."[8] Identifying the potential that resources such as GIS can offer the film historian, scholars such as Allen are therefore pushing forward research in this area in new, significant, and productive ways. Allen's project Going to the Show uses over 750 Sanborn Fire Insurance maps of 45 towns and cities in North Carolina between 1896 and 1922. Drawing on a dataset featuring information on 1,300 movie venues identified from the maps and an extensive archive of contextual materials, such as newspaper advertisements and articles, photographs, architectural drawings, and city directories, Going to the Show "situates early movie going within the experience of urban life in the state's big cities and small towns. It highlights the ways that race conditioned the experience of movie going for all North Carolinians – white, African American, and American Indian."[9] Similarly, Klenotic is using GIS technology to explore the social and geographic contexts of film distribution and exhibition in New Hampshire.[10] Both Allen and Klenotic are part of the HoMER network (History of Moviegoing, Exhibition and Reception), an international group of film scholars established in 2004 whose aim is to "promote understanding of the complex, international phenomena of filmgoing, exhibition, and reception."[11] The development and subsequent availability of database collations of data relating to film practices – whether in terms of production and exhibition, or geographies of consumption and location, or information on genre, studio locations, and patterns of distribution – has meant that cartographic methods of geohistorical analysis are now increasingly recognized as valuable tools for historical research on film.[12]

The second category of "cinematic cartography," film-related tourism, represents an area that has developed into something of a global phenomenon in recent years, prompting the publication not only of movie maps of cities and regions (often as tie-ins to major film releases) but also of a string of film-tourism travel guides that have become a focus of research into movie-mapping and place-marketing. The British Tourism Authority's (BTA) *Movie Map of Britain* (1990) was the first national campaign that sought to capitalize on the economic potential of film-related tourism; the map became BTA's (now Visit Britain) most successful printed product. The organization has since gone on to produce a series of movie maps and has become a global player in the film tourism market. Working with film production and distribution companies, Visit Britain has developed dedicated film tourism offices in Los Angeles and Mumbai and typically plans its location maps with movie studios at least twelve months in advance of the date of a major film release.[13] Film tourism has brought with it growing convergence between the film and tourism industries, with each providing mutually reinforcing promotional tie-ins and product or brand awareness designed to stimulate both the consumption of place (the economic imperative of the tourism, leisure, and cultural industries) and the consumption of film and television productions. A number of recent studies have examined the economic impacts or potential of this form of destination marketing, although studies that address the social, cultural, and geographic impacts of film-related tourism remain comparatively underdeveloped.[14]

By way of contrast, recent approaches in film studies have focused on the ways in which maps are embedded in feature films; as Tom Conley points out, "Since the advent of narrative in cinema – which is to say, from its very beginnings – maps are inserted in the field of the image to indicate where action 'takes place.'"[15] In Sébastien Caquard's discussion of cinematic maps – or "cinemaps" – he argues that early animated maps in films such as Fritz Lang's *M* (1931) predated many of the future functions of modern digital cartography, such as the use of sound, shifts in perspectives, and the combination of realistic images and cartographic symbols. Caquard suggests that professional cartographers can learn much from the study of cinematic techniques used by Lang and other filmmakers in terms of their status as cinematic precursors to modern forms of mediated cartography.[16] Conley also examines the use of maps in feature films focusing on examples from postwar cinema.[17] Conley's approach to what he terms "cartographic cinema" can be defined in terms of its focus on the geographic and representational cartographies contained within the diegesis and on the psychological and affective forms of mapping that are mobilized between film and viewer in terms of his or her subjectivity and psychic positionality.

Similarly focusing on feature films, Bruno's *Atlas of Emotions* provides a detailed theoretical exposition of the ways in which the affective properties of the cinematic medium play host to mappings of the psychic and emotional topographies that are given form in the immaterial architectures that structure the complex interplay between spatial textualities of film and the subjective "navigation" of these spaces by the viewer and spectator. For Bruno, the psychogeographic mobilities and affective geometries that are unleashed by film and other forms of moving image culture prompt renewed critical understanding not only of the ways we might read or "map" the spaces of film but also of how the forms and architectures of urban space might shape theoretical, aesthetic, and practical reengagements with cities themselves: "Mapping is the shared terrain in which the architectural-filmic bond resides – a terrain that can be fleshed out by rethinking practices of cartography for travelling cultures, with an awareness of the inscription of emotion within this motion. Indeed, by way of filmic representation, geography itself is being transformed and (e)mobilized. . . . A frame for cultural mappings, film is *modern cartography.*"[18] Conley's cartographic cinema treads a similar theoretical terrain to that of Bruno, noting that even if a film does not feature a map as part of its narrative, "by nature [film] bears an implicit relation with cartography. . . . Films *are* maps insofar as each medium can be defined as a form of what cartographers call 'locational media.'"[19] In a similar vein, Teresa Castro's discussion of the "mapping impulse" refers to a "visual regime," a way of seeing the world that has cartographic affinities.[20] Cinematic cartography here refers less to the presence of maps per se in films than to the cultural, perceptual, and cognitive processes that inform understandings of place and space. Focusing on what she describes as "cartographic shapes," Castro argues that "panoramas" (viewpoints shaping synoptic and spatially coherent landscapes and vistas), "atlases" (visual archives and spatiovisual assemblies), and "aerial views" ("god's-eye" or bird's-eye perspectives from planes or hot-air balloons) define a cinematic topography in which the mapping impulse is a central cognitive element. Drawing attention to the broad and complex theoretical terrain within which mapping and cartographic practices are embedded, Castro notes that "mapping can therefore refer to a multitude of processes, from the cognitive operations implied in the structuring of spatial knowledge to the discursive implications of a particular visual regime."[21]

Finally, the last category in Roberts's suggested five-point typology of cinematic cartographies is what the artist and filmmaker Patrick Keiller has dubbed "film as spatial critique."[22] To date, the most productive resource for research in this area has been archival film materials from the early days of film (1890s–1910s) and the postwar period (1950s–1970s). In the case of the latter, archival research into Liverpool on film illustrates ways in which a spatial reading of films of postwar urban landscapes exposes and articulates some of the contested or contradictory spatialities emerging during this period as a result of large-scale and controversial modernist urban planning, which left its destructive stamp on many cities during the 1960s and 1970s.[23]

In contrast to the emphasis on feature films in all the above approaches, the University of Liverpool's City in Film project is the first attempt to comprehensively map the wide range of genres and production practices that have contributed to how the local and regional has been perceived and projected in moving image media.[24] Focusing on factual productions, over 1,700 items have been cataloged, ranging from actualities, travelogues, newsreel footage, amateur and independent productions, promotional material, and campaign videos and enabling in-depth analysis and the development of a socially and spatially embedded reading of what Roberts has termed "the archive city."[25] As well as cataloging the films using conventional data such as title, producer, date, duration, format, and so on, wherever possible the films have been viewed and cataloged according to the buildings and locations depicted and their spatial function and use. This has enabled us to study in some depth the dynamic ways in which moving images are invested in the everyday production of locality, space, and subjective identities and how particular genres engage with a historically contingent geography of place. Transferring the City in Film catalog to a GIS platform enables the development of a more refined process of geohistorical analysis as well as the production of a range of georeferenced contextual materials, including digitized segments of particular films, interviews with filmmakers, cine society programmes, company/organizational material, and supporting documentation. The use of GIS also enables the research team to situate the content of films listed in the catalog using digital mapping tools, informing understandings of the ways in which the city is visualized by specific genres and at particular times. Furthermore, the mapping process is enabling a fruitful dialogic tension to emerge between different perceptions of place and identity as they are articulated in specific genres and filmmaking practices.

LOCATING THE CITY IN FILM

The use of the name Merseyside to depict a regional area raises the first problem that this research had to confront, that of the blurred and shifting boundaries between the city and its hinterland. Using Ordnance Survey maps from the 1890s to the present and historical boundary data, we can trace how the changing political and administrative boundaries of the city and its hinterland have shaped the cinematic geography of the films at different times. One of the questions that quickly emerge when mapping a city's representation in film is where to draw the boundaries that define the urban area. What or where is the object that is the "city in film"? At the start of our research the area bounded by the inner ring road to the east that demarcates the inner core of the nineteenth-century city from its twentieth-century suburban and industrial hinterland (Queens Drive) and the natural border of the River Mersey to the west and south marked out the area that was initially to be the geographical focus of enquiry. This was in large part a practical consideration – the need to delimit the amount of potential research material. However, as the current City in Film map of Liverpool and Merseyside shows, attempts to "geographically fix" a clearly defined urban spatial area have proved problematic, an indication that many of what can or could be regarded as "Liverpool films" takes us, by default, to consider a more